Inflation Inequality Across the Euro Area

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What We Do

- Inflation inequality across households increases when relative-price shocks interact with heterogeneous household baskets.
- We construct monthly indices by income, age, and residence for all 20 euro-area countries, calibrated to official HICP weights.
- We simulate the effects of temporary VAT cuts, gas and electricity price measures, and fuel rebates on average inflation and inflation inequality.

Inflation Inequality and Policy Effects, 2021–2023

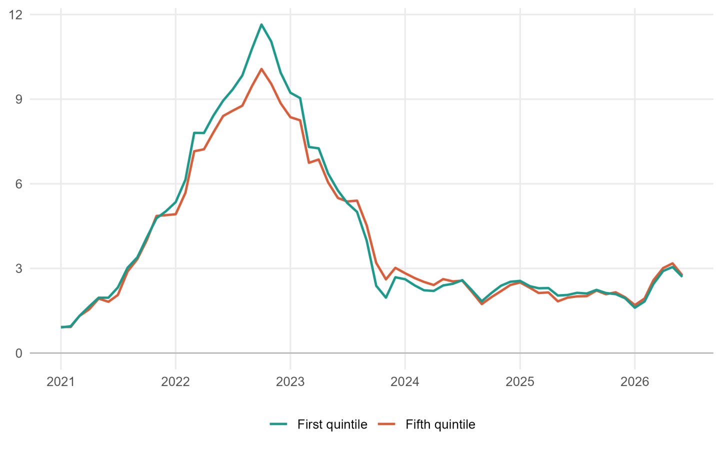
- Euro-area Q1–Q5 inflation gap rose to **0.9 point**, from 0.1 point in 2021, and returned to zero in 2023.
- Household energy contributed **1.6 points** to the 2022 gap; fuel, transport, and other products offset most of this exposure.
- Household price policies lowered annual inflation by **1.3 points in 2022** and **0.6 point in 2023**, while narrowing the Q1–Q5 gap by **0.4** and **0.5 point**.

Linking Official Prices to Household Baskets

- We link monthly Eurostat HICP prices and official item weights to HBS expenditure shares through a harmonized COICOP bridge.
- HBS tabulations identify baskets by income, age, and residence; microdata add household expenditures, disposable income, and demographics.
- We use RAS to calculate expenditure weights that simultaneously satisfy two margins: the expenditure share of each household category and the official HICP weight of each product.

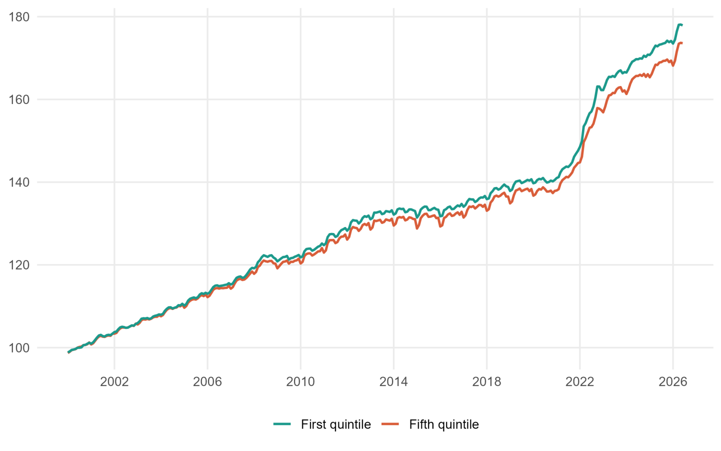
- Interpretation: baskets are held fixed between HBS waves, so the baseline abstracts from substitution and within-group heterogeneity.

Low-Income Households Faced Higher Inflation During the Energy Shock



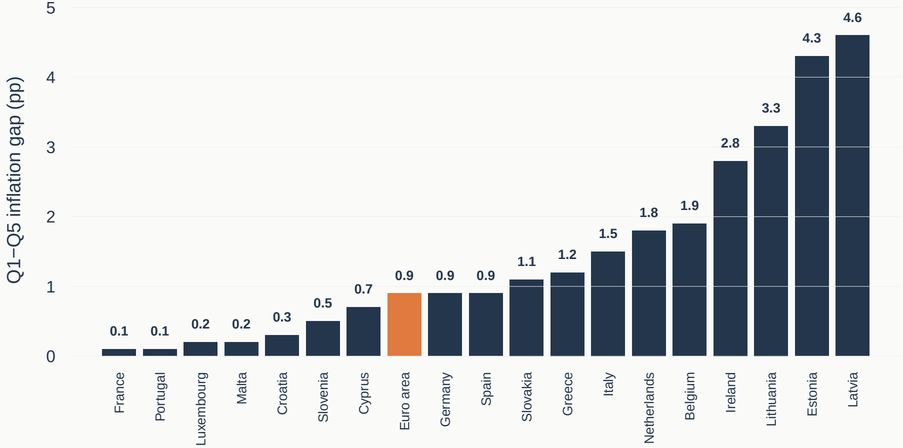
Source: Eurostat HICP and HBS; authors' calculations. Units: year-on-year inflation, in percent.

Since 2000, Only the 2021–23 Shock Had a Sizable Effect on Inflation Inequality



Source: Eurostat HICP and HBS; authors' calculations. Units: price index, 2000=100.

Q1–Q5 Inflation Gaps Across Euro-Area Countries in 2022



Source: Eurostat HICP and HBS; authors' calculations. Countries with a negative Q1–Q5 gap in 2022 are excluded; the euro-area aggregate is shown as a benchmark.

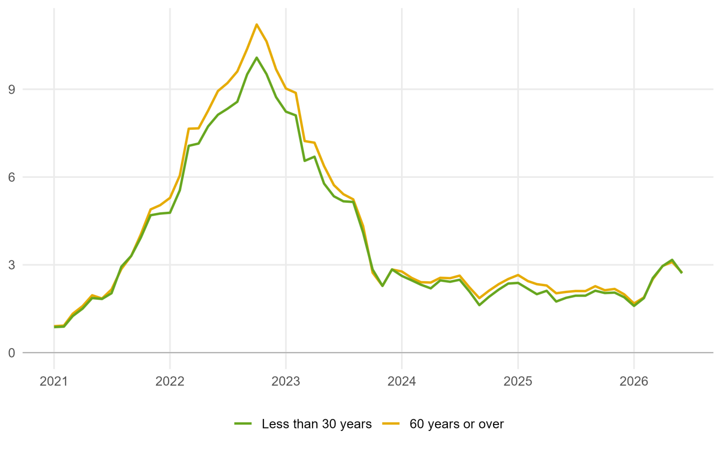
Household Energy Drove the 2022 Gap; Other Products Offset It

Product group	Q1–Q5 inequality contribution		
	2021	**2022**	2023
Food and non-alcoholic beverages	0.1	0.6	0.6
Housing energy	0.4	**1.6**	0.1
Operation of personal transport equipment	-0.2	-0.4	0.0
Transport (excluding vehicle operation)	-0.1	-0.3	-0.2
Other	-0.1	-1.0	-0.5
Total	**0.1**	**0.9**	**0.0**

Food kept widening the gap in 2023, but offsets elsewhere brought the aggregate gap back to zero.

Source: Eurostat HICP and HBS; authors' calculations. Product rows are contributions to the Q1–Q5 annual inflation-rate gap, in percentage points.

Older Households Faced Higher Inflation



Source: Eurostat HICP and HBS; authors' calculations. Units: year-on-year inflation, in percent.

Household-Level Price Indices

- We treat each household as an elementary group. Its individual weight $\omega_{j,i,2020}^{\text{HBS}}$ combines its expenditure on item j with the official HICP weight for that item.

$$P_{i,y,m} = \sum_{j \in \mathcal{J}_{i,2020,y}} \omega_{j,i,2020}^{\text{HBS}} \frac{p_{j,y,m}}{p_{j,y,0}}.$$

- Individual indices measure inflation heterogeneity within income, age, and residence categories, which broad group averages conceal.
- This within-category dispersion was substantial during the 2021–2023 inflation surge.

Household Groups Hide Household-Level Dispersion

Average inflation and dispersion were both higher in 2022.

fig/fig_EA20_household_inflation_distribution_2021_20

Source: Eurostat HICP and HBS microdata; authors' calculations. Units: annual household inflation (x) and household share (y), in percent.

Three Instruments Lowered Retail Energy Prices

- The counterfactual covers **190 price measures** across 20 countries, costing **EUR 268.9 billion**; **88%** of the cost was untargeted.
- **Price caps and tariff shields** limited household electricity and gas prices.
  France: gas tariffs were frozen and the 2022 electricity increase was capped at 4%.
- **Temporary VAT and excise-tax reductions** lowered consumer energy prices.
- **Fuel rebates and regulated discounts** reduced prices at the pump.
  France: the fuel rebate rose from EUR 0.18 to EUR 0.30 per litre in autumn 2022.

Source: National policy sources; Sgaravatti et al.; authors' calculations.

Defining the Price-Policy Counterfactual

- The counterfactual removes these price-based interventions component by component while retaining observed prices for unaffected products.

Example: reversing a temporary VAT cut

$$\widetilde{p}_{j,t} = p_{j,t}\frac{1+\tau^0_{j,t}}{1+\tau^1_{j,t}}, \qquad \tau^0_{j,t} = \text{VAT rate}, \quad \tau^1_{j,t} = \text{temporary VAT rate.}$$

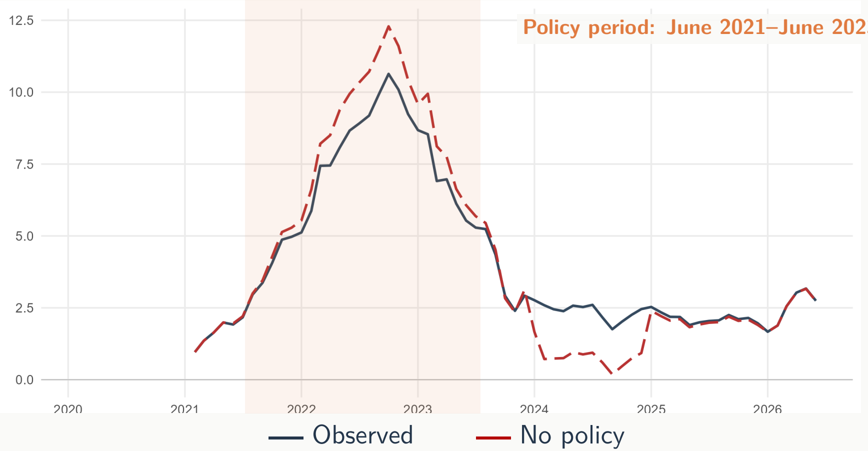
*Belgium: cutting VAT from **21%** to **6%** implies*

$$\widetilde{p}_{j,t} = p_{j,t}\frac{1.21}{1.06} \simeq 1.142\, p_{j,t}.$$

$j =$ electricity and $t =$ March 2022–December 2023.

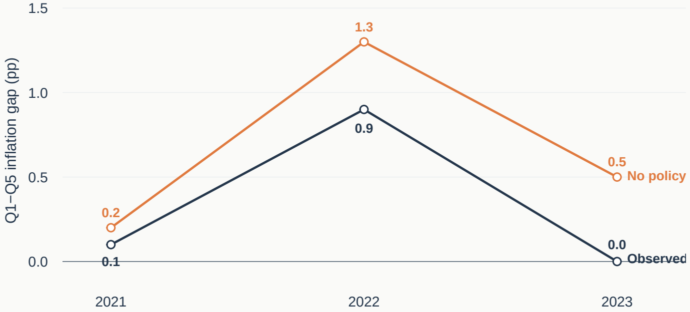
Source: National policy sources, Sgaravatti et al.; authors' calculations.

Price Policies Compressed the Inflation Peak



Source: Eurostat and national statistical institutes; authors' calculations. Units: year-on-year inflation, in percent; January 2021–July 2026.

Policies Cut the Gap by 0.4 pp in 2022 and 0.5 pp in 2023



Policy effect on the gap = Observed gap − No-policy gap

2022: $0.9 - 1.3 = -\mathbf{0.4}$ pp **2023:** $0.0 - 0.5 = -\mathbf{0.5}$ pp

Source: Eurostat and national statistical institutes; authors' calculations. Annual-average Q1–Q5 inflation-rate gaps, in percentage points.

The Inflation Gap Was Temporary; Policy Cushioning Persisted

- Q1–Q5 inflation inequality rose from **0.1 point in 2021** to **0.9 point in 2022**, then returned to zero in 2023; national experiences remained far more dispersed.
- Household energy contributed **1.6 points** to the 2022 gap, while fuel, transport, and other consumption categories offset it.
- Price policies reduced annual inflation by **1.3 points in 2022** and **0.6 point in 2023**; they reduced inequality by **0.4** and **0.5 point**.
- Administered household-energy prices generated most of the distributional cushioning; fuel support was mildly regressive in 2022.

Appendix

Policy Effects Differed Sharply Across Countries and Years

Country	Average-inflation effect			Q1–Q5 inequality effect		
	2021	2022	2023	2021	2022	2023
Germany	0.0	-0.5	-0.7	0.0	-0.2	-0.7
France	-0.1	-2.2	-1.9	0.0	-0.9	-1.1
Italy	-0.2	-1.3	0.7	-0.1	-0.1	-0.1
Spain	-0.4	-2.0	-0.4	-0.2	-0.7	-0.4
Netherlands	0.0	-1.6	-0.7	0.0	-0.5	-0.6
Portugal	0.0	-0.8	-1.4	0.0	-0.3	-0.9
Slovakia	0.0	0.0	-3.0	0.0	0.0	-1.9
Euro area	**-0.1**	**-1.3**	**-0.6**	**-0.1**	**-0.4**	**-0.5**

- Negative effects indicate that unconventional fiscal policies reduced inflation or the Q1–Q5 gap.

Source: Eurostat and national statistical institutes; authors' calculations. Entries are observed minus no-policy annual-average inflation rates, in percentage points.

Rural Households Faced Higher Inflation at the Peak

Source: Eurostat HICP and HBS; authors' calculations. Units: year-on-year inflation, in percent.

A Uniform Energy Shock Mostly Raises Average Inflation, Not the Gap

Product group	Average inflation	Q1–Q5 gap	Direct gap	Indirect gap
Food	0.7	0.1	0.0	0.1
Housing energy	2.9	0.9	0.7	0.3
Fuel	1.4	-0.6	-0.5	-0.1
Transport excluding fuel	0.9	-0.1	0.0	-0.1
Other products	1.7	-0.1	0.0	-0.1
Total	**7.6**	**0.2**	**0.1**	**0.1**

- Housing energy widens the gap by **0.9 point**, but greater Q5 exposure to fuel and transport offsets most of it.

Source: Eurostat HICP, HBS and 2022 FIGARO; authors' calculations. Simulation of a uniform 40% energy-sector shock; percentage-point effects.

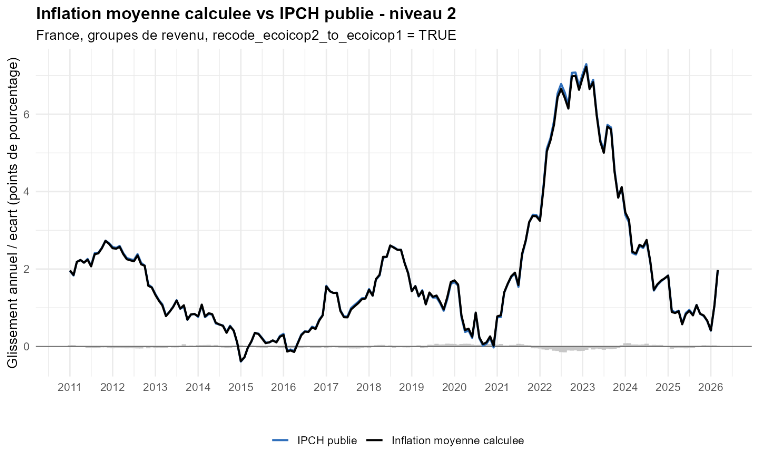
Household-Energy Measures Account for the Inequality Reduction

Product group	Average-inflation effect 2021	2022	2023	Q1–Q5 inequality effect 2021	2022	2023
Food and non-alcoholic beverages	0.0	0.0	-0.1	0.0	0.0	0.0
Housing energy	-0.1	-1.0	-0.8	-0.1	-0.5	-0.5
Fuel	0.0	-0.3	0.3	0.0	0.1	0.0
All other products	0.0	0.0	0.0	0.0	0.0	0.0
Total	**-0.1**	**-1.3**	**-0.6**	**-0.1**	**-0.4**	**-0.5**

- Household energy lowers the gap by **0.5 point** in both 2022 and 2023; the 2022 fuel effect offsets 0.1 point.

Source: Eurostat and national statistical institutes; authors' calculations. Product contributions to annual policy effects, in percentage points.

RAS-Calibrated Inflation Tracks the Published HICP



Source: INSEE, Eurostat; authors' calculations. Units: year-on-year inflation, in percent.

Inflation Burden Is Larger for Low-Income Households

- The Q1 burden reached **3.1%, 10.2%, and 7.4%** of net income in 2021–2023, versus **2.0%, 5.7%, and 4.1%** for Q5.

Source: Eurostat HICP and HBS microdata; authors' calculations. Units: annual inflation burden, in percent of household net income.

The Paper Isolates the Household Expenditure Channel

- **Expenditure:** households buy different baskets, so relative-price shocks generate different inflation rates. This is the channel measured here.
- **Income:** wages, pensions, and transfers adjust at different speeds, producing unequal changes in real disposable income.
- **Wealth:** inflation erodes nominal assets and liabilities, transferring wealth from creditors to debtors.
- Our indices capture heterogeneity in expenditure shares within common COICOP price changes.
- They do not capture household-specific prices paid, product quality, within-category substitution, or the income and wealth channels.

Source: Main text; presentation framework adapted from the seminar PowerPoint.

Inflation Inequality Was Highly Uneven Across Countries

Country	2021	2022	2023	Country	2021	2022	2023
Austria	-0.5	-0.2	0.5	Ireland	0.4	2.8	1.1
Belgium	1.2	1.9	-2.3	Italy	0.1	1.5	0.4
Croatia	0.0	0.3	-0.3	Latvia	-0.2	4.6	2.4
Cyprus	0.0	0.7	0.2	Lithuania	0.2	3.3	0.4
Estonia	0.5	4.3	1.5	Luxembourg	0.1	0.2	-0.1
Finland	-0.2	-1.1	0.2	Malta	-0.4	0.2	1.2
France	0.3	0.1	0.1	Netherlands	-0.3	1.8	-0.9
Germany	-0.4	0.9	0.4	Portugal	0.7	0.1	-0.9
Greece	0.3	1.2	-0.5	Slovakia	-0.9	1.1	1.3
Spain	0.8	0.9	-0.7	Slovenia	0.1	0.5	1.4
Euro area					**0.1**	**0.9**	**0.0**

Source: Eurostat HICP and HBS; authors' calculations. Units: Q1 minus Q5 annual-average inflation rates, in percentage points.

Administered Prices Drove the Distributional Effect

Policy category	Fiscal cost EUR bn	Average-inflation effect 2021	2022	2023	Q1–Q5 inequality effect 2021	2022	2023
VAT cuts	63.3	0.0	-0.1	-0.1	0.0	0.0	0.0
Administered prices	163.8	0.0	-0.5	-0.8	0.0	-0.3	-0.5
Fuel subsidies and rebates	17.0	0.0	-0.3	0.3	0.0	0.1	-0.1
Other calibrated layer	16.3	0.0	-0.3	0.1	0.0	-0.2	0.0
Unallocated costs	8.5	–	–	–	–	–	–
Total	**268.9**	**-0.1**	**-1.2**	**-0.5**	**-0.1**	**-0.4**	**-0.5**
Untargeted measures	236.2	-0.1	-0.8	-0.1	0.0	-0.2	-0.3
Targeted measures	32.8	0.0	-0.5	-0.4	0.0	-0.2	-0.2

Negative values indicate that policies reduced inflation or the inflation gap.

Source: Bruegel; national sources; authors' calculations. Annual policy effects are observed minus no-policy rates, in percentage points. Category effects need not sum because policies interact.